Spatial–Spectral Attention Pyramid Network for Hyperspectral Stripe Restoration

Liangliang Chen, Yueming Wang, and Chengkang Zhang

***Abstract*— Push-broom hyperspectral imaging systems often suffer from stripe artifacts. The conventional methods treat the artifacts as noise and suppress narrow-stripe ones well but show limitations to wide and full-band stripe artifacts. To address the problem, this article proposes a spatial–spectral attention pyramid network (SAPN) for hyperspectral stripe restoration. First, the spatial–spectral mixed attention (SMA) module is developed to tackle the inefficiency of neighborhood representation in wide stripes, and it is mainly composed of three spatial and spectral attention (SSA) operations. Each SSA specifically combines channel and nonlocal attention (NLA) to compute spatial–spectral attention features. SMA utilizes these SSA operations to achieve different spatial–spectral attention features for multidirectional slices of hyperspectral cubes and then fuses them to establish the contextual connection between the single pixel and the global information. Furthermore, we build an efficient pyramid backbone (EPB) for stripe restoration. In EPB, multiresolution shallow pyramid features are extracted by the lightweight head module and then inferred and reconstructed by SMA and other layers from coarse to fine, and the shareable SSA layer also greatly decreases parameters. Besides, we develop an unsupervised learning strategy where SAPN generates pseudo-reference images with the aid of deep image prior (DIP) and achieve the convergent model for batch images. Experiments are carried out on the private and public hyperspectral datasets where wide stripes exist at the same and different spatial locations in all bands. Experimental results demonstrate that SAPN can obtain competitive objective metrics, and it can restore images with more realistic texture and fidelity spectra.**

***Index Terms*— Hyperspectral image, image restoration, mixed attention, wide stripe.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain rich spectral information, and they provide more easily distinguishable features to identify ground object for remote sensing. The push-broom imaging system can obtain high-resolution HSIs, but it easily suffers from stripe-shaped image artifacts because of the limitations of imaging equipment and its surroundings [1]. Image quality is the basic guarantee for the effectiveness of hyperspectral remote sensing applications. However, the stripe artifacts destroy the integrity of spatial and spectral information in HSIs, leading to degradation of HSI quality. Thus, research on restoring hyperspectral stripes is of great significance for improving remote sensing applications.

Hyperspectral stripe artifact includes the stripe effect of inconsistent radiometric calibration [2], [3], the deadlines of missing pixels [4], and other stripe-shaped pixel damage. Here, we focus on the wide-stripe-shaped invalid pixels collected by the hyperspectral detectors with machine malfunction or unsatisfactory environment, such as horizontal black stripes in Fig. 1(a). This kind of extreme stripe damage is not a general trouble, but particularly severely degrades image quality. Common stripes are narrow and randomly distributed throughout the image, and they are usually regarded as special noises with the characteristics of clear angle and stripe shape. Therefore, denoising-based methods are applied to reconstruct HSIs with stripe artifacts. According to different assumptions and priors, traditional destriping methods are usually divided into statistical and filter-based methods, optimization-based methods, and combination of these techniques [5]. Statistical methods adjust stripe pixels based on the similarity between subscenes but ignore the irregularity of stripe distribution, such as histogram modification and equalization methods. Filter-based methods consider stripes to be separable components in the original signal, For example, some customized filtering algorithms are proposed to implement destriping for specialized imaging devices in [1] and [6]. Lu et al. [7] and Li et al. [8] transformed the image signal into the sparse domain by jointly utilizing the sparse representation of spatial–spectral features and then removed the components of stripe noise. This kind of methods focus on the study of signal decomposition in specific transform domain but can easily lead to blurring or ringing artifacts due to its inappropriate signal removal. The optimization methods treat the restoration of hyperspectral stripe as an ill-conditioned problem and construct denoising optimization models based on various prior knowledge. The methods of [9] and [10] complemented the regularization constraint on the sparsity of HSIs to build total variation models. According to the low-rank property along the spectral dimension, some methods [4], [11], [12] built the low-rank matrix or tensor models to improve the fidelity of spectral features. However, optimization models for the ill-conditioned

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problem reconstruct images rather slowly because of their high computational complexity, and their strict assumptions also lead to poor robustness. The deep learning method driven by a large amount of prior data performs excellently on various vision tasks, and thus, it is popularly applied in hyperspectral denoising tasks [13]. Some methods [14], [15], [16], [17] construct deep convolutional networks (residual network and generative adversarial network) to remove stripe noise and other noise, in which specific operations, such as multiscale convolution and multichannel filtering, improve the ability of feature representation and perception. Furthermore, Zhang et al. [15] and Yuan et al. [17] proposed the spatial–spectral feature extraction and the spectral-aware loss function based on the intercorrelation prior of spatial and spectral features in HSIs. For the denoising-based method, small-area noisy pixels are restored thanks to intact neighbor pixels. The denoising-based methods can repair narrow stripes (1 or 2 pixels wide) well but fail to work for wide-stripe damage due to the lack of effective reference information from neighboring pixels, as shown in Fig. 1(b) [18]. Aiming at this problem, some image stripe restoration methods regard removing stripe damage (deadlines) as image inpainting. For example, the methods in [19], [20], [21], [22], and [23] utilize the interspectral correlation and the spatial nonlocal self-similarity to estimate the normal information of stripe pixels. Imperfectly, these methods are only suitable for partial-band stripe damage and perform poorly for full-band stripe damage, as shown in Case1 of Fig. 1(a), because they fail to obtain the spatial information from other intact bands, while the methods in [24], [25], [26], and [27] predicted and filled the stripe-shaped holes according to the data distribution consistency between the damaged and intact areas and even the methods in [28], [29], [30], and [31] used the depth image prior to achieve unsupervised HSI restoration. Importantly, the ability of data distribution representation and inference directly affects the reconstructed image quality. For example, Yu et al. [26] introduced texture attention to improve feature representation and inference of spatial details, and Liu et al. [27] used the partial convolution to enhance the robustness to stripe locations. Miao et al. [29] developed the discrete spatial–spectral deep prior based on the deep hyperspectral image prior (HSDIP) [28] and improved the feature representation of the restoration network. Niresi and Chi [30] introduced the constraint of least favorable distribution to enhance the perception of deep image prior (DIP). However, for each HSI, the learning framework of DIPs requires much time to complete iteration and convergence. Also, the reproducibility of image restoration requires saving the input values and model parameters for each image. In addition, the wide stripes across all bands weaken the intrinsic relationship of spatial–spectral features, making it difficult to effectively infer the missing pixels, such as the blurred artifacts of the inpainting-based method in Fig. 1(b) [32].

To address these problems, this article proposes a spatial–spectral attention pyramid network (SAPN) for hyperspectral stripe restoration. Inspired by the experience that narrow stripes are easier to repair, an efficient pyramid backbone (EPB) is built to restore the HSI stripe from coarse to fine. In the network, the spatial–spectral mixed attention (SMA) fully mines multidirectional contextual information to infer the pixels that lose spatial and spectral neighbors in wide and full-bands stripes. Furthermore, we develop an unsupervised restoration strategy that facilitates the reproduction of image stripe restoration. Finally, we perform HSI restoration for wide-stripe artifacts with the same and different spatial locations on private and public datasets. Experimental results show that our method can perform better in terms of objective metrics, spatial texture clarity, and spectral fidelity.

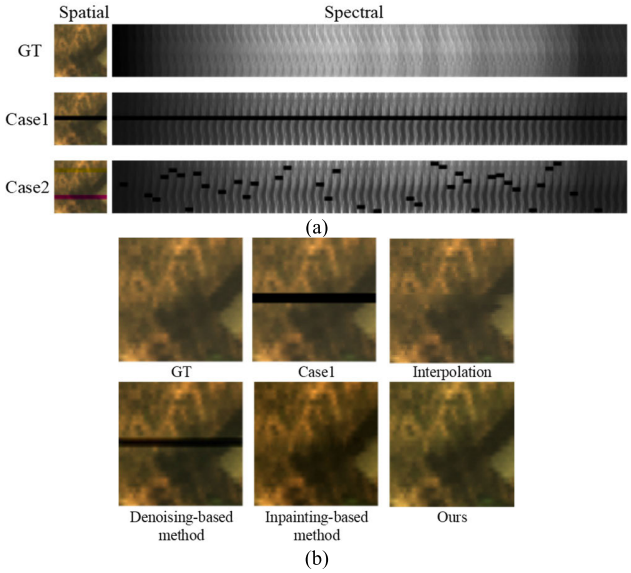


Fig. 1. Hyperspectral stripe damage and its restoration results. (a) GT is the ground-truth image, and Case1 and Case2 represent the wide-stripe damage with the same and different spatial locations, respectively. (b) Restoration results of the interpolation method, the denoising-based method [18], the inpainting-based method [32], and our method.

The main contributions of this article are as follows.

1) It develops an efficient pyramid network for hyperspectral stripe restoration. The network downsamples the wide-stripe image to reduce the negative impact of missing neighbors, and then, it progressively inpaints and upsamples the downsampled image. Also, the lightweight pyramid feature extraction module enhances the multiresolution feature representation ability. Compared to reconstructing directly from random noise, the network can reconstruct images from coarse to fine and achieve more realistic results with fewer parameters.
2) The SMA module is proposed, and it can compute spatial–spectral feature attention along three orthogonal directions to obtain more contextual priors. The multidirectional priors establish tight relationships for pixels with few neighbors. In the module, the attention feature at a certain position is squeezed and expanded to flexibly balance its feature representation capability and computational complexity. The module significantly improves the feature inference ability of the restoration network for wide and full-band stripes.
3) The unsupervised restoration strategy is proposed to overcome the shortcomings of DIP. In the unsupervised strategy, the network learns the shareable prior knowledge of mapping from the stripe image to the ground

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truth, rather than iteratively computing random noise. This strategy enables a converged network to easily repair multiple historical and new samples without the huge storage and time overhead.

The remainder is organized as follows. Section II introduces deep learning for hyperspectral destriping, DIPs, and attention mechanisms. The details of the proposed method are presented in Section III. Section IV shows the experiments and their analysis. Section V gives the conclusions of the work.

II. Related Works

A. Deep Learning for Hyperspectral Destriping

Recently, destriping methods based on deep learning have attracted extensive attention. For example, Guan et al. [33] utilized recurrent network to mine intraband and interband correlations and achieved stripe restoration. Wang et al. [34] computed the unbiased estimation loss of nonpure images and developed a semisupervised deep convolutional network for hyperspectral destriping. These methods regard stripes as special noise, and their essence is to remove components consistent with the noise prior or obtain pure components from the existing image signal. Different from the above denoising-based methods, the inpainting-based method considers that the information of stripe pixels is destroyed or lost, so it usually estimates stripe pixels based on existing image content. For example, Lai et al. [35] combined the optimization-based method with the deep learning one to develop deep plug-and-play prior for hyperspectral image restoration (DPHSIR), and it leveraged DIP to construct the optimization problem of estimating intact images. Rasti et al. [31] proposed a method of unsupervised deep hyperspectral inpainting (DeepHyIn) using a new mixing model and the image priors of abundance sum and nonnegative constraints. Ma et al. [32] developed the combination of intrinsic image characterization with robust noise modeling (GLIC_NM) and explored the global correlation and local regularization of HSIs, as well as the intrinsic properties of stripe artifacts, and then applied a posterior probabilistic network to estimate damaged or missing pixels. Niresi and Chi [3] used the framework of DIP combined with differentiable regularization of data rank to inpaint damaged and missing pixels (RDLRHyIn). The wide stripes here suffer from the loss of many pixels, and the denoising-based methods inefficiently remove the stripe components or extract pure and complete signals. Instead, it prefers to estimate the corresponding intact pixels by the inpainting-based methods.

B. Deep Image Prior

DIP presents outstanding performance for image restoration, and it states that the convolutional neural network (CNN) can restore the damaged image only by iterative computing random noise [36]. Generally, CNN needs to learn prior knowledge from a large number of training samples and reach a convergence state. Differently, in the DIP method, the random Gaussian noise is the input of CNN, and the uncorrupted pixels constraint its iterative calculation process. In essence, the final convergence CNN represents the mapping relationship between the data distribution of random Gaussian and that of the undamaged pixels. The computing process of DIP for image restoration is shown in Table I.

TABLE I
Process of DIP

Algorithm 1 The Process of Deep Image Prior

ε: random Gaussian tensor; f_N: the image restoration network; θ_t: the network parameters calculated for the t-th iteration.
n: the number of iterative calculations; x_{mask}: the mask of damaged pixels; x_{stp}: the damaged image; l_{mse}: the loss function of mean square error

(1) Randomly initialize the input ε and the image restoration network $f_N(\cdot;\theta)$, and its parameters $\theta = \theta_0$.
(2) for $t = 1, \ldots, n$;
(3) ε is input in f_N, and obtain the image x_t:
$$x_t = f_N(\varepsilon;\theta).$$
(4) Calculate the loss value l_{dip}:
$$l_{dip} = l_{mse}\left(x_t \cdot x_{mask},\ x_{stp} \cdot x_{mask}\right).$$
(5) l_{dip} performs Back Propagation to obtain θ_{t+1}.
(6) The optimizer updates the network parameters:
$$\theta = \theta_{t+1}$$
(7) End for
(8) Obtain the restoration image: x_n

In HSIs, local pixels have a certain distribution law, that is, the context information of pixels is correlated. Therefore, the DIP method is also successfully applied to HSI restoration [28]. Based on the DIP framework, Miao et al. [29] used a linear mixture model to separate spatial and spectral information and established the independent inference subnetworks for spectral and spatial features. Although work [29] improves hyperspectral denoising, it also exhibits limitations on spectral fidelity because of ignoring the relationship between spectral and spatial information. Niresi and Chi [30] introduced the least favorable distribution to design the Huber loss function and its excellent feature perception accurately guides the inference of the target data distribution. Zhuang et al. [37] constrained the iterative computation of DIP via low-rank and sparse priors as well as plug-and-play priors to unsupervisedly reconstruct the noise-free image. In [38], spatial–spectral local smoothness prior and sparse noise prior were explored to constrain the solution of the optimization model. Therefore, similar to the other CNN methods, the DIP method is also affected by its feature representation and inference ability. However, in the damaged image, the wide-stripe area obviously has a different distribution from the intact area, and there are many pixels without neighbors in wide stripes, which negatively affects the feature representation and inference of the context prior. Also, the DIP method performs iterative calculations for individual image restoration. This restoration process makes it impossible to reconstruct images quickly without saving the input and the converged network for each image. To address these issues, this article proposes an SAPN for hyperspectral stripe restoration.

C. Attention Mechanism of HSI

The attention mechanism can well enhance the feature representation and reasoning ability of the network. Currently, most attention applications are variations of classic squeeze-and-excitation attention (SEA) [39] and nonlocal

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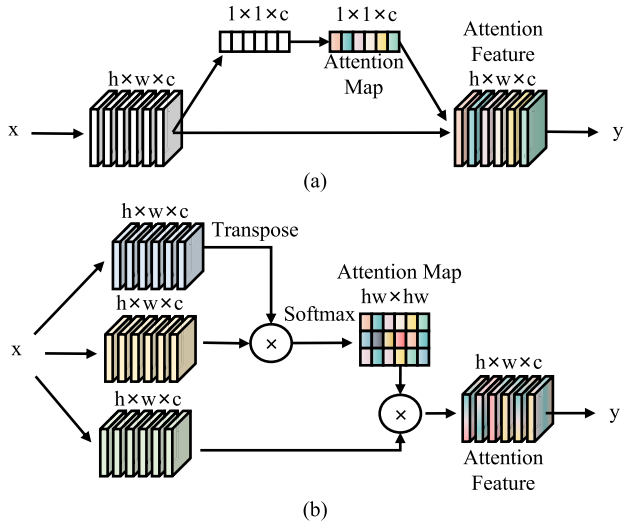


Fig. 2. SEA and NLA. (a) and (b) Inference progress of SEA and NLA, respectively. In (a) and (b), "x" is the input, "y" is the output, "$h \times w \times c$" represents the size of features, "Transpose" is the transpose operation, and "Softmax" is the Softmax function.

attention (NLA) [40]. SEA and NLA are shown in Fig. 2(a) and (b), respectively. From Fig. 2, we can see that SEA can calculate the channel attention map to adjust each channel of the feature and improve the saliency of useful channels. The channel attention map implies the context information between channels. Differently, the attention map of NLA represents the connection between pixels in the global spatial domain. For example, Shi et al. [41] developed a 3-D attention denoising network, where spectral and spatial attention features were inferred by their separate NLA branches. Yang et al. [42] proposed a spatial–spectral joint attention network, and it cascaded different deformations of SEA to calculate spectral and spatial attention features in turn. Pan et al. [43] built a quasi-attention recurrent unit, in which the defined quasi-recurrent pooling function can obtain the correlations of spatial features in the spectral domain, and the NLA of the spectral bands strengthens the correlations of adjacent spectra. For wide stripes, the lack of extensive neighborhood information motivates the need of broader contextual features. However, spatial–spectral attention computed in independent branches or cascades discounts the correlation of spatial features across spectral bands. Quasi-attention relies heavily on spectral context information but is slightly insufficient for the Case1 stripes where the full spectrum information is lost. Therefore, we propose an SMA module, and it mixes NLA in SEA to calculate the joint contextual connection of the spatial and spectral domains and obtains the SMA features, avoiding information loss caused by excessive compression of dimensions. Also, SMA calculates and fuses multidirectional attention features to suppress the negative impact caused by the completely information loss in a certain dimension.

III. Method

This article proposes an SAPN for hyperspectral stripe restoration. In the network, an EPB is built for coarse-to-fine feature inference, and it consists of two parts: *Head* and *Infer*. The *Head* part efficiently extracts multiresolution features, and the *Infer* part is responsible for feature inference and image reconstruction. In the *Infer* part, an SMA module is developed to represent the pixelwise correlation and fully mines the dependence of contextual information. Furthermore, this article explores an unsupervised restoration strategy to learn the mapping from the damaged image to its ground truth. The details of SAPN are shown in Fig. 3.

A. SMA Module

Different types of HIS stripes are shown in Fig. 4, and they are distinguished by the spatial position of stripes. In the "*Case1*" HSI, some pixels of wide stripes contain partial spatial and full spectral invalid neighborhoods, as shown in Fig. 4(a). Therefore, it is difficult for the "*Case1*" HSI to reconstruct the pixels of wide stripes, especially the spectral information. Compared to the "*Case1*" stripe, the "*Case2*" stripe is randomly distributed in each band. Therefore, in the image with the "*Case2*" stripe, fewer pixels lose all spectral information, but more pixels suffer from the damage of local spectral information, as shown in Fig. 4(b). More damaged pixels of the "*Case2*" stripe negatively affect spatial feature representation and reconstruction of HSI.

To address this problem, we propose the SMA module, and its detailed structure is shown in Fig. 3. The calculation process of SMA consists of the inference operations of different spatial and spectral attention (SSA) and the fusion operation, as shown in Fig. 5.

In the plane perpendicular to the spectral direction (denoted as the *HW* plane), we compress the spectral features of each spatial pixel rather than directly using its statistical features (average or max values), so as to obtain more accurate contextual relations. The SSA feature X_{hw} of the *HW* plane is calculated as follows:

$$\begin{aligned} X_{hw} &= f_{hw}(X_{HW\times C}) \\ &= f_{\text{upw}}\left(f_{\text{sa}}\left(f_{\text{pw}}(X_{HW\times C})\right)\right) \end{aligned} \tag{1}$$

where $X_{HW\times C}$ is the reshaped feature tensor with size of $H \cdot W \times C$, and f_{pw}, f_{sa}, and f_{upw} represent the operations of compressing the spectral features, calculating the NLA [40], and expanding the spectral features, respectively. In SSA of Fig. 3, f_{pw} corresponds to the left Linear+ReLU, f_{upw} corresponds to the right Linear+softmax, and f_{sa} corresponds to the layers between f_{pw} and f_{upw}. In detail, $X_{HW\times C}$ is compressed into the feature with size of $H \cdot W/r_w \times C$ by f_{pw}, and then, f_{sa} computes it to obtain the spatial–spectral attention features with size of $H \cdot W/r_w \times C$, and its NLA map with size of $(H \cdot W/r_w) \times (H \cdot W/r_w)$ represents the pixelwise correlation. Finally, f_{upw} expands the compressed SSA feature to the one of size $H \cdot W \times C$. The value of r_w is negatively correlated with the computational complexity, but too large r_w also leads to excessive feature compression and performance degradation. Here, we set r_w to 4 according to the experiment.

Considering that the NLA map of the *HW* plane only contains the spatial nonlocal correlation, those of the *CW* plane and the *CH* plane are calculated to obtain diverse nonlocal

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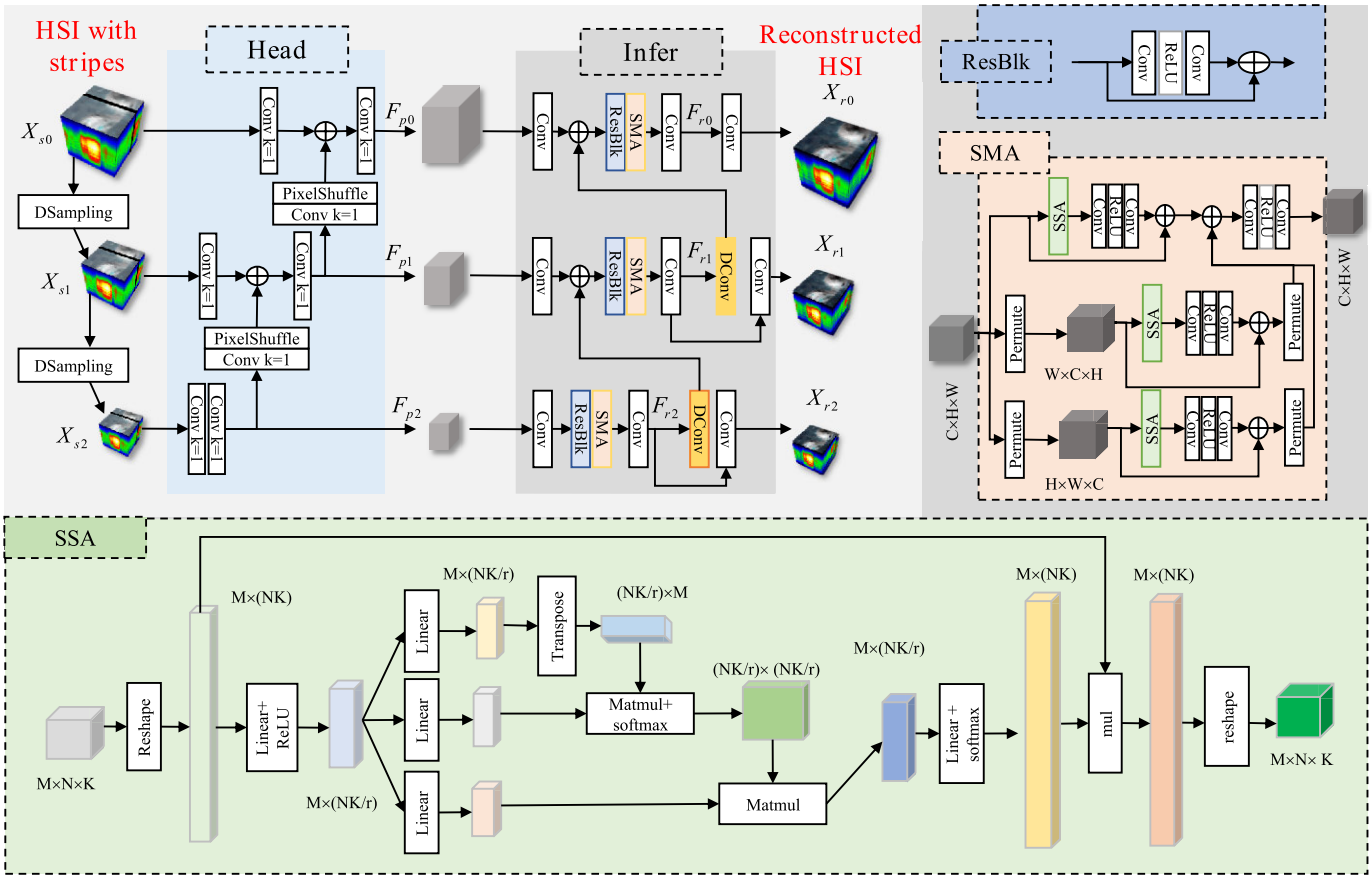


Fig. 3. Structure of SAPN. "Conv" and "Conv $k = 1$" are the convolution layer with 3×3 and 1×1 kernels, respectively. "PixelShuffle" is the periodic shuffling operator [34] and "DConv" is the deconvolution layer, and they both perform the upsampling operation. "ReLU" and "softmax" are the activate layers, "SSA" is the spatial–spectral attention, "Reshape" rearranges the dimension of a tensor without changing its elements, "Permute" represents the transpose operation of tensors to align with other tensors, "Linear" is the 1-D convolution, and "Matmul" and "mul" are the matmul product and the Hadamard product, respectively. "$C \times H \times W$" or "$M \times N \times K$" represents the size of data.

correlations of the spectral information. Similar to f_{hw}, the calculation formulas of f_{wc} and f_{ch} are as follows:

$$\begin{aligned} X_{wc} &= f_{wc}(X_{WC\times H}) \\ &= f_{\text{upc}}\left(f_{\text{sa}}\left(f_{\text{pc}}(X_{WC\times H})\right)\right) \qquad (2) \\ X_{ch} &= f_{ch}(X_{CH\times W}) \\ &= f_{\text{uph}}\left(f_{\text{sa}}\left(f_{ph}(X_{CH\times W})\right)\right) \qquad (3) \end{aligned}$$

where f_{pw} and f_{upw} represent the operations of compressing and expanding features along the C direction, respectively, and f_{ph} and f_{uph} are ones along the H-direction. The nonlocal maps of size $(W \cdot C/r_c) \times (W \cdot C/r_c)$ and $(C \cdot H/r_h) \times (C \cdot H/r_h)$ represent different correlations of spatial and spectral features, and thus, X_{ch} and X_{wc} can provide more spectral priors for the *Case1* stripe restoration and spatial priors for the *Case2* stripe restoration.

X_{hw}, X_{ch}, and X_{wc} contain multidirectional features of spatial and spectral information, and they complement each other. Therefore, we fuse X_{hw}, X_{ch}, and X_{wc} to obtain the SMA features X_{SMA}, formulated as

$$X_{\text{SMA}} = f_{\text{mix}}(X_{hw}, X_{ch}, X_{wc}) \qquad (4)$$

where f_{mix} is a feature fusion operation and its detailed structure is shown in SMA of Fig. 3.

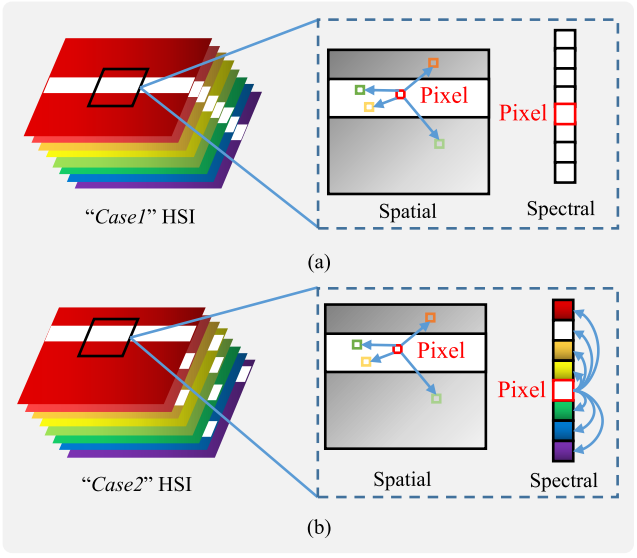


Fig. 4. HSI stripe. (a) and (b) Wide stripes with same and different spatial positions in all bands, respectively. In (a), "*Case1*" HSI represents the HSI where white indicates missing pixels, and right is the spatial and spectral images of the black rectangle area. The same is true for (b).

In X_{SMA}, the complementary spatial and spectral features guarantee reliable priors for pixels without neighborhood. Different from spatial–spectral features in 3-D convolution

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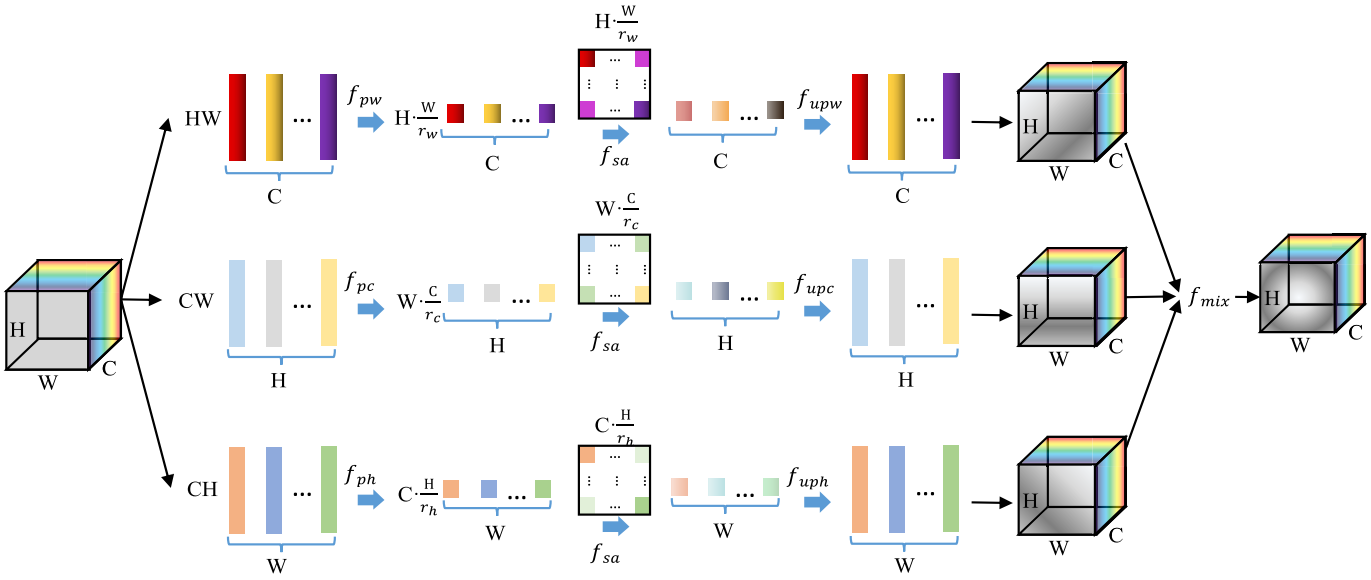


Fig. 5. Calculation process of SMA. C, H, and W are, respectively, the spectral bands, the spatial height, and the width of hyperspectral features, and their compression factors correspond to r_c, r_h, and r_w (here, $r_c = r_h = r_w = 4$). f_{sa} is the NLA operation [40], and f_{pw} and f_{upw} are the feature compression and expansion operations of $X_{HW\times C}$, respectively. Similarly, f_{pc} and f_{upc} are those of $X_{WC\times H}$, and f_{ph} and f_{uph} are those of $X_{CH\times W}$. f_{mix} is the fusion operation.

[44] and the deep spatial–spectral global reasoning network (GRN) [45], X_{SMA} acquires spatial–spectral joint attention in *HW*, *WC*, and *CH* planes with 2-D operations and avoids the separation of spatial and spectral information. Also, X_{hw}, X_{ch}, and X_{wc} use the compressed features instead of traditional statistical values as pixel features of NLA, improving feature representation of X_{SMA}. For convenience and lightweight of SAPN, X_{hw}, X_{ch}, and X_{wc} share the same structure of SSA module, and their structures are shown in Fig. 3. Note that r_c, r_h, and r_w are adjusted to ensure that feature dimensions match f_{sa}.

B. Efficient Pyramid Backbone

The EPB consists of two parts: *Head* and *Infer*. The *Head* layer extracts shallow features from multiresolution images, and the *Infer* layer is responsible for feature inference to obtain the repaired image. The structure of EPB is shown in Fig. 3.

The images X_{s0}, X_{s1}, and X_{s2} are input into the *Head* layer, and they are downsampled from the stripe-damage image by nearest neighbor interpolation (the downsampling factors are set to 1, 2, and 4, respectively). According to the structure of *Head* in Fig. 3, F_{p1} is the fusion of X_{s1} and X_{s2}. Similarly, F_{p0} contains the information from X_{s0}, X_{s1}, and X_{s2}. The method of [46] demonstrates that multiresolution features can improve the inference results. Differently, *Head* further performs multiresolution fusion with compact operations where the convolution is set to the 1×1 kernel and the periodic shuffling operator [47] upsamples the low-resolution features, and it negligibly increases the parameters of EPB.

The *Infer* layer, as the main body of EPB, is responsible for multiresolution feature inference and reconstruction. Fig. 6 presents the schematic of progressive image restoration (Plan A) and one-step image restoration (Plan B). In general, narrower stripes are easier to repair. As shown in Plan A of Fig. 6, the convolution calculation with the 3×3 kernel or interpolation can repair the single-pixel wide stripe. However, for Plan B, the convolution operation with 3×3 kernels has a small receptive field and fails to perceive effective information of the wide stripe. The convolution with large kernels greatly increases the computational overhead, and the expansion of the solution space also slows down the training convergence. Therefore, we employ low-resolution image restoration and upsampling to accomplish coarse-to-fine feature inference. As shown in Fig. 3, the multiresolution features from *Head* are input into *Infer*. In the feature inference process, F_{p2} is first deduced through the *conv*, *ResBlk*, and *SMA* units to obtain F_{r2}. Next, F_{r1} is obtained by inferring F_{p1} and F_{p2}. Different from F_{r2}, F_{r1} additionally absorbs the upsampling feature from the deconvolution of F_{p2}. Essentially, F_{r1} is the result of super-resolution reconstruction performed on the restored low-resolution feature F_{r2}, and F_{p1} provides it with high-resolution auxiliary features. Similarly, F_{r0} is inferred from F_{r1} and F_{p0}. Furthermore, F_{r0}, F_{r1}, and F_{r2} are, respectively, reconstructed into HSIs X_{r0}, X_{r1}, and X_{r2}. Image priors with different resolutions can constrain the coarse-to-fine reconstruction process by the loss function l_{ms}, formulated as

$$l_{\text{ms}} = l_{\text{mse}}\left(X_{r0}, X_{g0}\right) + w_1 l_{\text{mse}}\left(X_{r1}, X_{g1}\right) + w_2 l_{\text{mse}}\left(X_{r2}, X_{g2}\right) \tag{5}$$

where X_{g0}, X_{g1}, and X_{g2} are, respectively, the downsampled ground truth with the scale factor of 1, 2, and 4, and l_{mse} is the loss function of mean square error. Here, the weight coefficients w_1 and w_2 are set to the empirical value of 0.5.

C. Unsupervised Image Restoration

The image in-painting methods based on DIP can realize the hyperspectral stripe restoration through unsupervised

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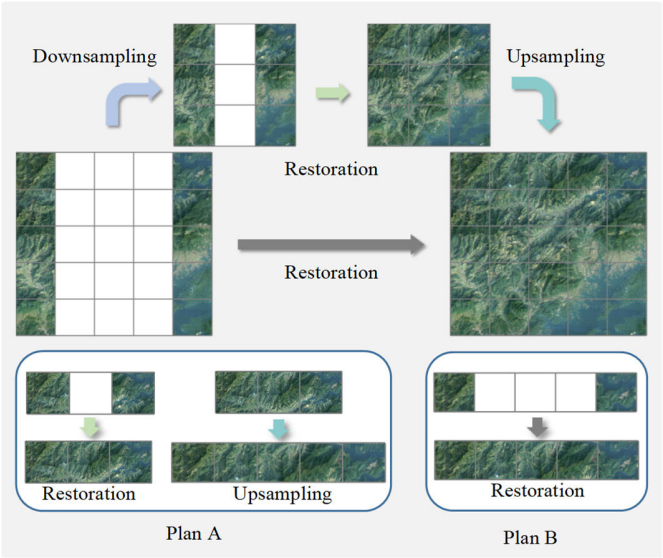


Fig. 6. Schematic of different image restoration methods. Plan A represents progressive image restoration, and it includes the operations of narrow-stripe restoration and upsampling. Plan B is one-step image restoration, and it directly restores the lost pixels.

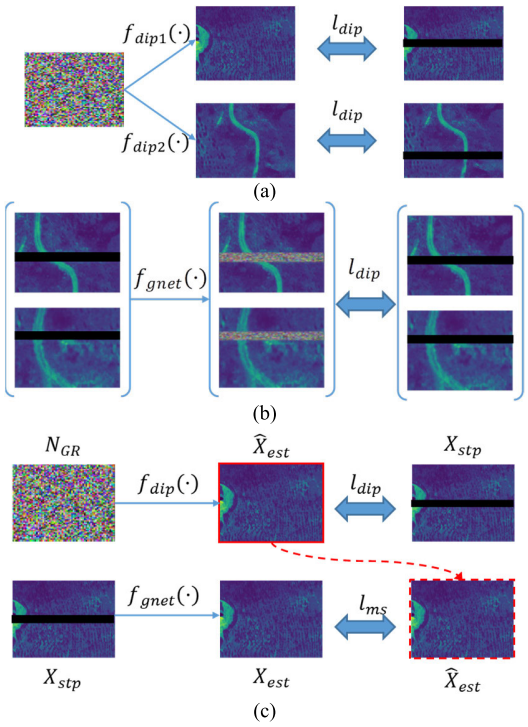


Fig. 7. Network learning with different inputs. (a) DIP method whose input is random noise, (b) one whose input is the damaged image, and (c) proposed unsupervised learning strategy. l_{dip} is presented in Table I, and l_{ms} is formulated as (5). f_{dip1} and f_{dip2} are networks with the same structure but different prior knowledge of images. f_{gnet} is the network trained on multiple images. The red arrow indicates $\hat{X}_{\text{est}}$ reconstructed by the converged f_{dip} as the reference image for l_{ms}.

learning, but its convergent network is only valid for its corresponding sample and not for other samples. To address this problem, this article proposes an unsupervised image restoration based on DIP and obtains a general network for all samples.

According to [36], DIP obtains a specific mapping from the input random noise to the target image, and its unshareable prior knowledge is limiting for DIP to batch image restoration. As shown in Fig. 7(a), the two images need to be iteratively trained separately to obtain their respective convergence models f_{dip1} and f_{dip2}. On the other hand, the image restoration network performs training on all damaged images, and its loss function is constructed with constraints on intact pixels. However, this method fails for stripe restoration when the input is replaced with the damaged image, as shown in Fig. 7(b), because the convergent network f_{gnet} that only learns the mapping relationship between intact areas contains no mapping knowledge between striped areas, and the striped areas are different from the intact areas in terms of feature distribution. Inspired by the unsupervised image super-resolution, we use the reconstructed result of DIP as the simulated ground-truth label and develop the unsupervised learning strategy in Fig. 7(c). First, the random noise N_{GR} is input into f_{dip} and generates the simulated ground truth $\hat{X}_{\text{est}}$. Second, the network f_{gnet} of X_{stp} mapped to X_{est} is trained with the pyramid loss function l_{ms} where $\hat{X}_{\text{est}}$ is the simulated ground truth of X_{est}.

The simulated ground truth enables f_{gnet} to perform supervised learning for image restoration, and the converged f_{gnet} can be used for batch image restoration because of its shareable prior knowledge. Although the simulated ground truth is inferior to the real one, this method can achieve unsupervised image restoration utilizing a shared network and obtain the comparable results. Here, both f_{dip} and f_{gnet} adopt SAPN.

IV. Experiment

A. Experiment Datasets

1) Private Datasets: The dataset is a scene in the suburbs of Shanghai, collected by an airborne visible near-infrared (VNIR) hyperspectral camera at noon on November 14, 2020. It includes 63 spectral bands from 393 to 986 nm, and its size is 3000 × 3000 with the spatial resolution of 0.20 m. Here, we cut the dataset into several 64 × 64 × 63 patches and remove the ones with stripes. Finally, 1985 patches are used for experiments, which are divided into 1585 training patches and 400 testing patches.

2) Public Datasets: Pavia Centre dataset (Pavia) acquired by the Reflective Optics System Imaging Spectrometer sensor is a 1096 × 1096 image with 102 bands, and it can freely available from the website (https://www.ehu.eus/ccwintco/index.php/ Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University). Washington DC dataset acquired by the Hyperspectral Digital Imagery Collection Experiment sensor is a 1208 × 307 image with 191 bands, and it can be available from the website (https://engineering.purdue.edu/~biehl/ MultiSpec). Here, we select the part scene of Pavia with size of 192 × 192 × 80 and the part of Washington DC with size of 256 × 256 × 191 as the test datasets, as shown in [23].

3) Real Datasets: GaoFen-5 (GF5) dataset [48] is a shortwave infrared image with a size of 512 × 512 × 180 and a spatial resolution of 30 m, and it is acquired by the Advanced

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Hyperspectral Imager (AHSI) aboard China's GF5 satellite [49]. The bands of 42–58, 95–114, and 176–180 are water absorption bands and their information is seriously damaged. VNIR513 and VNIR1025 are the parts of private datasets with real-world wide stripes, and the image size is $512 \times 512 \times 63$. Differently, VNIR513 and VNIR1025 contain the real-world stripes of approximately 27 pixels wide, and the degradations are distributed in the same spatial locations of all bands.

For the above private dataset, we synthesize the artificial images by randomly adding 4-pixel-wide stripes and finally construct the 1585 pairs of training images and 400 pairs of testing images. Real wide-stripe pixels generally come from interpolation or copies of intact neighborhood pixels, but here, stripe areas are directly masked by deadlines for convenience, and their spatial positions are generated randomly. In the dataset, one pair of images contains the ground truth and its damaged images with two types of stripes: the stripes of all bands with the same spatial positions (denoted as the Case1 dataset) and the ones of different bands with various spatial positions (denoted as the Case2 dataset). The public Pavia and Washington DC also suffer from the above synthetic stripe artifacts, and they are used for testing. Real datasets are used for validation of inpainting real-world hyperspectral stripes, and they all contain the real-world degradations. Differently, the wide strips of GF5, corresponding to the Case2 stripes, appear in part of bands, but those of VNIR513 and VNIR1025, corresponding to the Case1 stripes, appear at the same spatial position of all bands.

B. Experimental Details

Experiments are performed on a platform with Intel Core i7-12700 CPU, NVIDIA A100 GPU (32 G), Ubuntu 20.04.4, and we adopt the Fast HSI inpainting (FastHyIn) method [23], HSDIP [28], HSI denoising with realistic data (HSIDwRD) [18], GRN [45], DPHSIR [35], DeepHyIn [31], GLIC_NM [32], RDLRHyIn [3], and the method of geometric low-rank tensor approximation (GLRTA) [48], [50] as comparison methods. Among these methods, HSIDwRD and GRN are the denoising-based methods, and the others are the inpainting-based methods. Besides, the supervised approaches HSIDwRD, DPHSIR, and GRN are retrained with the training dataset of SAPN. The Adam optimizer is used for training and it is set to $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\varepsilon = 10^{-8}$. The weights of loss function l_{ms} are set to $w_1 = w_2 = 0.5$.

For the compared methods, their parameters, flops, and runtime using CPU mode present the comparison of computational overhead. Also, we adopt the image quality metrics: peak signal-to-noise ratio (PSNR), structural similarity (SSIM), spectral angle mapper (SAM), and Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), and further propose a weighted normalization metric (WNM) for comprehensive evaluation and comparison. The WNM value of the ith method can be formulated as

$$m_w^{(i)} = n_{\text{psnr}}^{(i)} + n_{\text{ssim}}^{(i)} - n_{\text{sam}}^{(i)} - n_{\text{ergas}}^{(i)} \tag{6}$$

where $n_{\text{psnr}}^{(i)}$, $n_{\text{ssim}}^{(i)}$, $n_{\text{sam}}^{(i)}$, and $n_{\text{sam}}^{(i)}$ are the normalized PSNR, SSIM, SAM, and ERGAS values of the ith method, respectively. $m_w^{(i)}$ is positively correlated with image quality, and its value ranges from -2 to 2. $n_{\text{psnr}}^{(i)}$ can be formulated as

$$n_{\text{psnr}}^{(i)} = \frac{m_{\text{psnr}}^{(i)} - \min\left(M_{\text{psnr}}\right)}{\max\left(M_{\text{psnr}}\right) - \min\left(M_{\text{psnr}}\right)} \tag{7}$$

where $M_{\text{psnr}} = [m_{\text{psnr}}^{(i)} | i = 0, 1, \ldots, 8]$, $m_{\text{psnr}}^{(i)}$ is the PSNR value of the ith method. Similar to $n_{\text{psnr}}^{(i)}$, the normalized values $n_{\text{ssim}}^{(i)}$, $n_{\text{sam}}^{(i)}$, and $n_{\text{ergas}}^{(i)}$ can be calculated. PSNR and SSIM focus on image quality of spatial information, while SAM and ERGAS do on that of spectral information. Also, their conflicting scores make it difficult to judge overall restoration effects. WNM can give a composite score to a certain extent regarding spatial and spectral information. In particular, the WNM index of less than 0 indicates relatively poor quality in terms of both spatial and spectral information.

C. Performance Comparison and Analysis

The objective results of different restoration methods are shown in Table II in which "Input" represents the original damaged image without hyperspectral restoration. It can be seen that SAPN achieves the best values of PSNR, SSIM, and SAM on the Case1 dataset, and the competitive ones on the Case2 dataset. For the Case2 dataset, SAPN is only inferior to HSIDwRD and GLRTA. HSIDwRD better reconstructs the images with Case2 stripes but performs particularly poorly for images with Case1 stripes. GLRTA and RDLRHyIn also show significant performance degradation. This is because the images of Case1 stripes suffer from streaks at the same position in all bands and fail to refer to spatial information in other bands. For the Case1 dataset, although HSDIP and RDLRHyIn, as the methods based on DIP, outperform other comparative methods; overall, it is slightly inferior to our method SAPN for Case1 and Case2. In terms of details, most of the inpainting-based methods achieve better results than the denoising-based ones; in particular, the methods of HSDIP, RDLRHyIn, GLRTA, and SAPN far exceed HSIDwRD and GRN.

For their computational overheads, FastHyIn and GLRTA are methods based on low-rank sparse representations and tensor approximation and thus have almost few parameters, but GLRTA has a large time overhead. Due to the iterative computation, the methods of HSDIP, DPHSIR, DeepHyIn, RDLRHyIn, and GLIC_NM have a very high and uncertain time overhead although most of them have fewer parameters and flops. HSIDwRD, GRN, and SAPN are the end-to-end methods, and SAPN has less parameters than HSIDwRD but more than GRN. Also, SAPN significantly outperforms HSIDwRD and GRN in terms of flops and runtime. Therefore, SAPN can perform more efficiently in a comprehensive manner with guaranteed performance.

For more intuitive comparison, we present the visualization results from the test dataset of Case1 and Case2 (550.mat, 1148.mat, 1378.mat, and 1639.mat). Fig. 8 presents the ground-truth images and their stripe images, and their red, green, and blue channels correspond to the 29th, 17th, and 8th bands, respectively. Figs. 9 and 10 show the repaired results of the Case1 and Case2 stripes, respectively. From Fig. 9, we can

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TABLE II
COMPARISON METRICS OF DIFFERENT DESTRIPING METHODS ON THE CASE1 AND CASE2 DATASETS

Methods	Params (M)	Flops (G)	Time (s)	Case 1					Case 2				
				PSNR	SSIM	SAM	ERGAS	WNM	PSNR	SSIM	SAM	ERGAS	WNM
Input	/	/	/	8.7805	0.0000	0.2135	825.98	-2.000	8.6277	0.0001	0.2111	840.36	-1.984
FastHyIn	/	/	0.3273	18.4018	0.7605	0.1957	2.4065	0.1496	24.4374	0.9197	0.0985	2.4482	0.6939
HSDIP	3	2	*	41.9928	0.9154	*0.0105*	**0.0559**	*1.8710*	46.7957	0.9527	0.0104	0.1046	1.4719
HSIDwRD	24	151	1.7871	24.1944	0.8179	0.0438	0.6101	1.1079	*54.1917*	*0.9940*	*0.0059*	0.1861	*1.6433*
GRN	1	75	0.3457	21.7716	0.7601	0.0939	0.5046	0.7376	21.3151	0.4495	0.2145	2.4569	-0.366
DPHSIR	14	98	*	18.7182	0.7452	0.1470	2.4935	0.3775	24.4370	0.9253	0.0985	2.4484	0.6995
DeepHyIn	2	1	*	28.0094	0.6708	0.0346	0.2533	1.1058	28.3536	0.6923	0.0342	0.2413	0.8275
GLIC_NM	0.5	0.5	17.460	24.1168	0.8815	0.0995	2.4973	0.9011	24.4386	0.9258	0.0986	2.4660	0.6995
RDLRHyIn	3	2	*	*42.2192*	*0.9282*	0.0777	0.0719	1.5662	52.2231	0.9670	0.0590	*0.0389*	1.3389
GLRTA	/	/	7.2257	34.6396	0.8389	0.0269	0.1102	1.5052	**76.3998**	**0.9989**	**0.0008**	**0.0063**	**2.0000**
SAPN	14	46	0.0933	**44.4148**	**0.9548**	**0.0064**	*0.0700*	**2.0000**	50.0825	0.9811	0.0072	0.0586	1.5638

'/' represents the inaccessible data, and '*' indicates that the corresponding methods perform multiple iterative calculations and overhead a very large amount of time. Here the values of flops are calculated when the methods run in a single iteration with input size of 64×64×63, and the time is the runtime of each method using CPU mode. Values in bold are optimal results, and values in italics are suboptimal results. The same goes for the following tables.

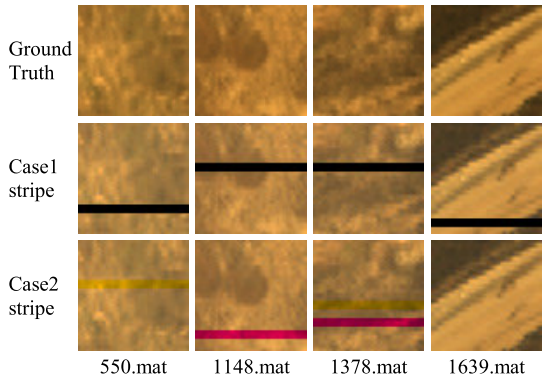


Fig. 8. Selected images from the ground-truth dataset. Images in the first row are their ground truth, and the ones in the second and third rows are their damaged images with the Case1 and Case2 stripes, respectively.

see that FastHyIn, HSDIP, GRN, and DPHSIR fail image restoration. DeepHyIn preserves some texture in nondamaged areas but obviously distorts the global radiance values. The inpainting areas of GLRTA are clearly distinguished from other areas. HSDIP, DeepHyIn, GLIC_NM, RDLRHyIn, and SAPN successfully repair the Case1 stripes, and the blur details and slight artifacts appear in the result of DeepHyIn. For the repaired results of the Case2 stripe in Fig. 10, GRN and DPHSIR achieve poor inpainting results, and DeepHyIn suppresses artifacts but still loses a lot of texture details. Combining the results of Figs. 9 and 10, only HSDIP, GLIC_NM, RDLRHyIn, and SAPN can consider the stripe damage of Case1 and Case2.

Furthermore, Fig. 11 presents the spectral curves of the repaired image "1639.mat" with the Case1 stripe and the image "1378.mat" with the Case2 stripe. The ordinate represents the normalized mean of all pixels in each band. From Fig. 11(a), the curves of g, h, and j maintain similar features with the curve ref of ground truth. In Fig. 11(b), the spectral curves are similar to ref except for curves e and g. Among them, curves h and j almost coincide with the ref. Comprehensively, SAPN can reconstruct more realistic spectral information on Case1 and Case2. In addition, the spectral distortion leads to the failure of objective metrics for the evaluation of reconstructed spatial features. For example, although Fig. 10(g) shows good results reconstructed by GLIC_NM, it still has poor objective metrics.

D. Generalization Comparison and Analysis

To compare the generalization of the destriping methods, we perform experiments on the public hyperspectral datasets DC and Pavia. Fig. 12 presents the ground-truth images and the damaged ones suffered from the Case1 and Case2 stripe artifacts, and their results of objective metrics, visualization images, and spectral curves are compared with each other.

Tables III and IV show the objective metrics on DC and Pavia, respectively. We can see that SAPN can achieve the best results in most indicators for restoring Case1 stripe damage. Also, the optimal WNM value also means that SAPN can best deal with the Case1 stripe as a whole. For the Case2 stripe restoration, the end-to-end method SAPN exhibits limitations due to its single forward computation, while the iterative approaches FastHyIn, DPHSIR, GLIC_NM, and GLRTA benefit from iteratively correcting errors based on neighborhood intact bands, significantly outperforming SAPN. Nonetheless, the advantage of iteration is ineffective for Case1 stripes due to the absence of intact bands, and these iterative approaches lag significantly behind SAPN for the Case1 stripe restoration. Other methods, including HSDIP, HSIDwRD, GRN, and DeepHyIn, lag behind SAPN in almost all objective metrics. Among them, HSIDwRD and GRN are significantly inferior to the other methods, which further demonstrates that the denoising-based methods difficultly cope with the wide-stripe damage. Besides, DeepHyIn and FastHyIn show greatly different performances on multiple datasets, and some methods further degrade image quality because of the

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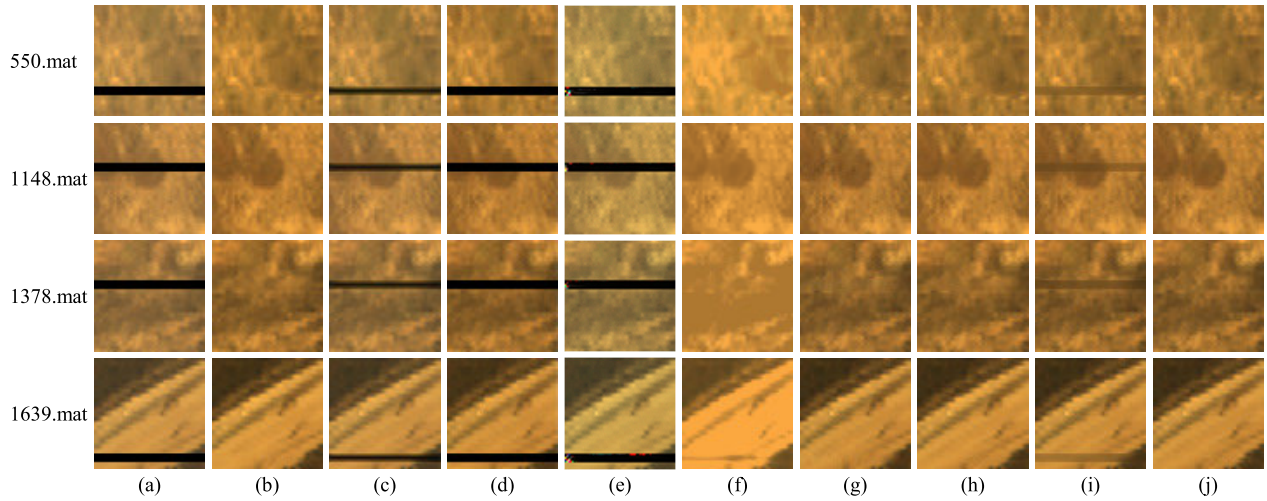


Fig. 9. Results of Case1 repaired with different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN.

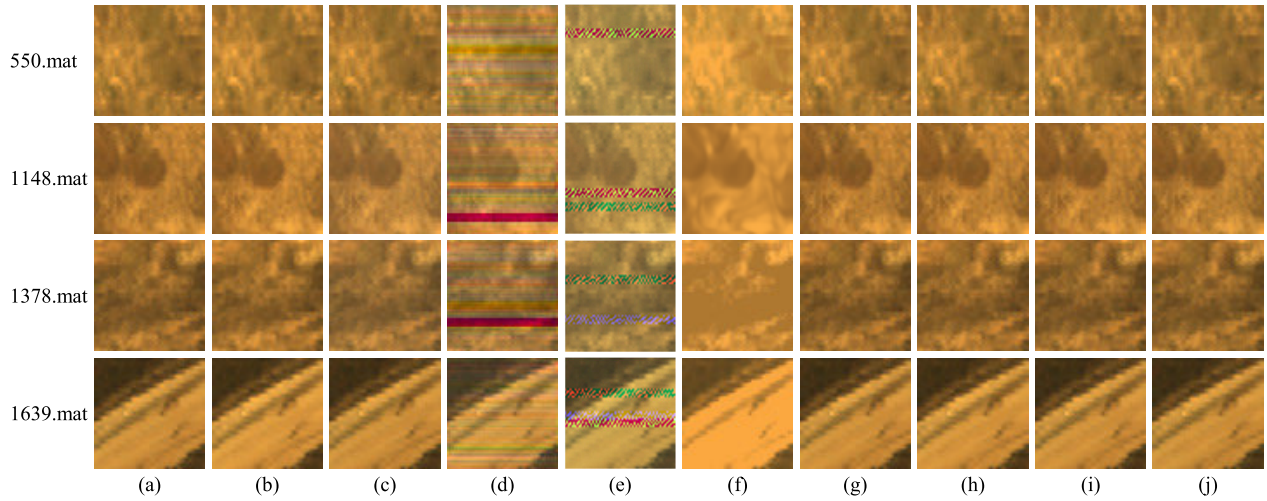


Fig. 10. Results of Case2 repaired with different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN.

destruction of intact areas, such as HSIDwRD, GRN, and RDLRHyIn. The unstable and further degraded performance illustrates the poor generalizability of the above methods. Finally, the comprehensive results of Tables III and IV prove that SAPN can better consider different wide-stripe restoration and achieve competitive objective metrics.

Figs. 13–16 present the visualized restoration results of DC and Pavia. It can be seen that GRN fails to inpainting both of the Case1 and Case2 stripes. FastHyIn and DPHSIR successfully repair the Case2 stripe but fail for the Case1 stripe. Serious radiation distortion appears in the results of GRN and DeepHyIn. From the results of the Case1 stripe restoration, we can see that HSDIP, HSIDwRD, GLIC_NM, and GLRTA obviously lose local details, and the wrong restored textures exist in Fig. 15(f). In contrast, SAPN can restore clear edges and textures for both Case1 and Case2 stripe artifacts.

Furthermore, the spectral curves of the restored pixels are shown in Fig. 17. We can see that HSIDwRD, GRN, and DeepHyIn suffer from significant spectral distortions on both DC and Pavia. This means that their objective metrics fail to evaluate the reconstruction results. For example, Tables III and IV show that HSIDwRD is better at repairing Case1 stripes compared to Case2 stripes, but its visual results in Figs. 13(c)–16(c) present the opposite conclusion. In detail, the curves of HSDIP and RDLRHyIn deviate slightly from that of the ground truth. DPHSIR, GLIC_NM, GLRTA, and SAPN can reconstruct fidelity spectral information on all images. Unfortunately, DPHSIR, GLIC_NM, and GLRTA lag behind SAPN in terms of spatial detail for the Case1 stripe.

In summary, the comprehensive results of objective metrics, visual images, and spectral curves indicate that SAPN can be better generalized to stripe restoration of other images, and it can restore clear textures and nearly real spectral information.

E. Experiments on Real HSIs

Furthermore, we performed the compared methods and SAPN on real-world datasets GF5, VNIR513, and VNIR1025. GF5 suffers from wide stripes in part of bands, similar to the Case2 stripe. VNIR513 and VNIR1025 contain the Case1

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TABLE III
COMPARISON METRICS OF DIFFERENT METHODS ON DC

Methods	Case 1					Case 2				
	PSNR	SSIM	SAM	ERGAS	WNM	PSNR	SSIM	SAM	ERGAS	WNM
Input	42.9181	0.9657	0.0307	0.0650	1.4511	43.6127	0.9665	0.1222	0.0606	0.7367
FastHyIn	42.9183	0.9657	0.0135	0.0650	1.4860	**140.611**	**1**	**0**	**0**	**2.0000**
HSDIP	45.7511	0.9196	0.0359	0.0495	1.4344	46.4415	0.9092	0.0395	0.0485	0.8614
HSIDwRD	29.1971	0.5113	0.3147	1.2585	-1.3354	29.1226	0.5148	0.3944	1.4102	-1.9672
GRN	19.9358	0.5528	0.4956	0.2309	-1.0291	28.0279	0.8027	0.2683	0.1043	-0.1376
DPHSIR	40.6691	0.9547	0.0133	0.1377	1.3486	64.9698	0.9805	0.0067	0.0557	1.2471
DeepHyIn	41.3118	0.7168	0.0990	0.0827	-0.4866	43.0293	0.8008	0.0818	0.0705	-0.5162
GLIC_NM	*53.4624*	**0.9867**	*0.0033*	*0.0201*	*1.8538*	*91.3909*	*0.9999*	0.0003	*0.0008*	*1.5715*
RDLRHyIn	37.4569	0.9767	0.0167	0.0285	1.3911	39.9765	0.9843	0.0181	0.0281	1.0288
GLRTA	41.3290	*0.9852*	**0.0029**	0.0193	1.5423	83.0482	*0.9999*	*0.0001*	*0.0001*	1.5001
SAPN	**59.0731**	0.9849	0.0149	**0.0176**	**1.9721**	59.4471	0.9850	0.0142	0.017	1.2169

TABLE IV
COMPARISON METRICS OF DIFFERENT METHODS ON PAVIA

Methods	Case 1					Case 2				
	PSNR	SSIM	SAM	ERGAS	WNM	PSNR	SSIM	SAM	ERGAS	WNM
Input	30.4241	0.9562	0.0409	0.0954	0.9393	30.8422	0.96	0.1349	0.087	0.6164
FastHyIn	30.4241	0.9562	0.0206	0.0953	1.0073	**126.114**	**1**	*0.0004*	**0**	**1.9994**
HSDIP	36.8784	0.9759	0.0272	0.0454	1.2538	41.1018	0.9954	0.0193	0.0279	1.1771
HSIDwRD	26.9895	0.8981	0.1792	0.1543	0.1986	25.6639	0.8869	0.2588	0.1761	-0.0050
GRN	25.3848	0.7276	0.3023	0.2021	-0.6028	26.2769	0.6931	0.3118	0.1989	-0.4465
DPHSIR	31.0637	0.9618	0.0140	0.0892	1.0637	77.6280	**1**	0.0020	0.0009	1.5773
DeepHyIn	25.3648	0.4775	0.1848	0.2048	-1.7175	26.1753	0.5399	0.1840	0.1965	-1.6402
GLIC_NM	*40.0082*	*0.9827*	**0.0036**	*0.0317*	*1.4379*	73.4022	**1**	0.0016	0.0008	1.5424
RDLRHyIn	37.6468	0.9780	0.0265	0.0416	1.2835	40.1037	0.9900	0.0284	0.0314	1.1290
GLRTA	39.3075	0.9811	0.0250	0.0327	1.4219	*84.1871*	*0.9999*	**0.0002**	*0.0001*	*1.6400*
SAPN	**61.8436**	**0.9998**	*0.0056*	**0.0027**	**1.9897**	49.1302	0.9966	0.0235	0.0120	1.2489

stripe. In these real images, the widest stripes exceed 20 pixels, and their inpainting results are shown in Figs. 18 and 19. With no ground-truth images, Fig. 18 only shows the repaired 45th band of GF5, and Fig. 19 presents the local stripe areas in the red box.

From Fig. 18, we can see that the results of FastHyIn, HSDIP, and HSIDwRD have severe image artifacts. GRN and GLIC_NM fail to reconstruct texture details. DeepHyIn loses some textures. GLRTA and SAPN exceed DPHSIR and RDLRHyIn in terms of the saliency of reconstructed details. For the repaired results of VNIR513 and VNIR1025, we can see that FastHyIn, HSIDwRD, GRN, DPHSIR, GLIC_NM, and GLRTA fail to repair such wide stripes, and HSDIP and DeepHyIn suffer distortions in contrast and radiation. The pixels repaired by RDLRHyIn do not correspond to the real situation. Compared with other methods, SAPN performs better in terms of radioactivity and texture authenticity.

Therefore, SAPN can also perform better than the comparison methods on the real datasets GF5, VNIR513, and VNIR1025.

F. Ablation Experiments for Key Modules

In order to verify the effectiveness of the key modules in SAPN, we construct its ablation networks (Net, Net+P, Net+P+SA, Net+P+SMA, and Net+P+TSMA) and perform them on the test sets of Case1 and Case2. Net+P+SMA represents SAPN where P and SMA represent the *Head* module and the SMA module in Fig. 3, respectively. In Net+P+SA and Net+P+TSMA, the symbol SA represents the modified SMA where the branches f_{cw} and f_{ch} are removed and the retained branch f_{hw} is set to $r_w = 1$; the symbol TSMA means that X_{hw}, X_{ch}, and X_{wc} use three separate SSA modules. Net+P is the simplified SAPN where the SMA modules are replaced with six residual blocks, and Net is the simplified Net+P where *Head* is replaced with two common convolution layers with 3×3 kernels. Table V shows the objective results of these ablation networks.

From Table V, we can see that Net+P can achieve the better PSNR and SSIM values but worse SAM and ERGAS values compared with Net. This shows that the *Head* module can improve the spatial features but lose the spectral

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TABLE V
OBJECTIVE METRICS OF ABLATION NETWORKS ON CASE1 AND CASE2

Methods	Parameters (M)	Case 1				Case 2			
		PSNR	SSIM	SAM	ERGAS	PSNR	SSIM	SAM	ERGAS
Net	3.80	42.9338	0.9431	0.0070	0.0774	49.0863	0.9745	0.0081	0.0718
Net+P	3.29	43.2330	0.9468	0.0080	0.0778	49.2782	0.9758	0.0078	0.0863
Net+P+SA	91.42	44.1633	0.9527	0.0068	0.0741	*50.1962*	0.9792	**0.0069**	0.0640
Net+P+SMA	13.60	**44.4148**	*0.9548*	**0.0064**	**0.0700**	**50.4922**	*0.9815*	*0.0071*	**0.0553**
Net+P+TSMA	135.47	*44.2167*	**0.9863**	*0.0065*	*0.0703*	49.8366	**0.9979**	0.0077	*0.0589*

The meanings of Net, P, SA, SMA, TSMA are introduced in *F. Ablation experiments for key modules.*

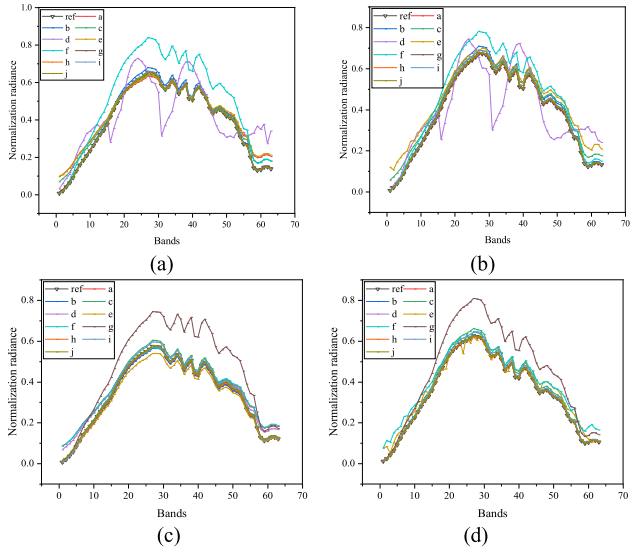


Fig. 11. Spectral curves of images reconstructed by different methods. (a)–(d) Curves of mean values per bands of "550.mat," "1148.mat," "1378.mat," and "1639.mat," respectively. Among them, the Case1 stripe appears in "550.mat" and "1148.mat," and the Case2 stripe does in "1378.mat" and "1639.mat." a–j respectively represent the methods of FastHyIn, HSDIP, HSIDwRD, GRN, DPHSIR, DeepHyIn, GLIC_NM, RDLRHyIn, GLRTA, and SAPN, respectively, and the ref represents the ground truth.

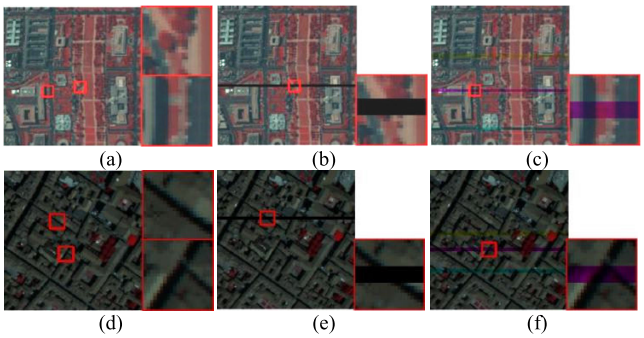


Fig. 12. DC and Pavia, and their damaged images with the Case1 and Case2 stripes. (a) Ground truth of DC and (b) and (c) DC images with the Case1 and Case2 stripes, respectively. (d) Ground truth of Pavia and (e) and (f) Pavia images with the Case1 and Case2 stripes, respectively.

information due to ignoring the spatial–spectral correlation. Also, the Head module has less parameters compared with the common convolution layers. Net+P+SA outperforms Net and Net+P in terms of all objective metrics, demonstrating that the spatial–spectral attention extracted by SSA can improve

TABLE VI
OBJECTIVE METRICS OF ATTENTION-BASED METHODS

Methods	Dataset	PSNR	SSIM	SAM	ERGAS
SAPN	Case1	**44.4148**	*0.9548*	**0.0064**	**0.0700**
SQRN		31.9910	0.9076	*0.0314*	0.5673
MAN		*39.2387*	**0.9700**	0.1335	*0.2383*
SAPN	Case2	**50.4922**	*0.9815*	**0.0071**	**0.0553**
SQRN		34.3451	0.9254	0.0547	0.4624
MAN		*43.0696*	**0.9890**	*0.0172*	*0.1443*

the feature inference of hyperspectral information. However, SA greatly increases the computational overhead. Furthermore, SMA enhances feature representation of spatial–spectral attention by collecting and fusing multidirectional attention and drives Net+P+SMA to achieve the better results in terms of spatial and spectral restoration with less parameters. Besides, Net+P+TSMA increases the parameters by two orders of magnitude but achieves comparable performance to Net+P+SMA.

Fig. 20 shows the visual results of these ablation networks on the hyperspectral image "1639.mat," and Fig. 20(a) and (b) shows its ground-truth image and damaged one, respectively. In their heat maps of the integral error, we can see that Net+P reduces the error of the damaged pixels and its surrounding pixels but increases that of the undamaged areas due to ignoring the spectral correlation. Compared with Net+P and Net, Net+P+SA attenuates distortion in both damaged and undamaged areas. Furthermore, less spatial–spectral distortion errors exist in the result of Net+P+SMA, but more ones do in that of Net+P+TSMA. This significant improvement shows that SA and SMA can effectively improve the feature fidelity of restored images, and SMA does better.

G. Comparison With Other Attention-Based Methods

Furthermore, two attention-based methods are added for comparison, and they are spatial–spectral quasi-attention recurrent network (SQRN) [43] and mixed attention network (MAN) [51], respectively. For intuitive comparison, SQRN and MAN are configured with more intermediate feature channels to achieve a comparable amount of parameters, and their objective and visual results are shown in Table VI and Fig. 20.

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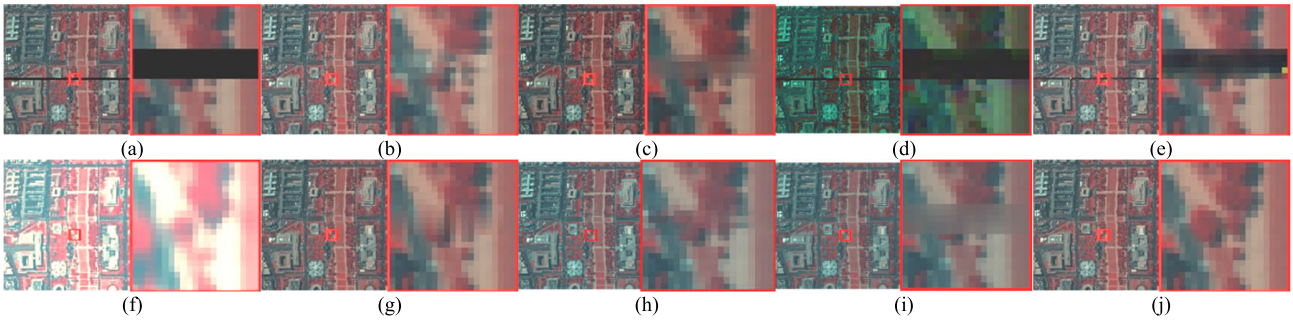


Fig. 13. Results of DC with the Case1 stripes repaired by different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN.

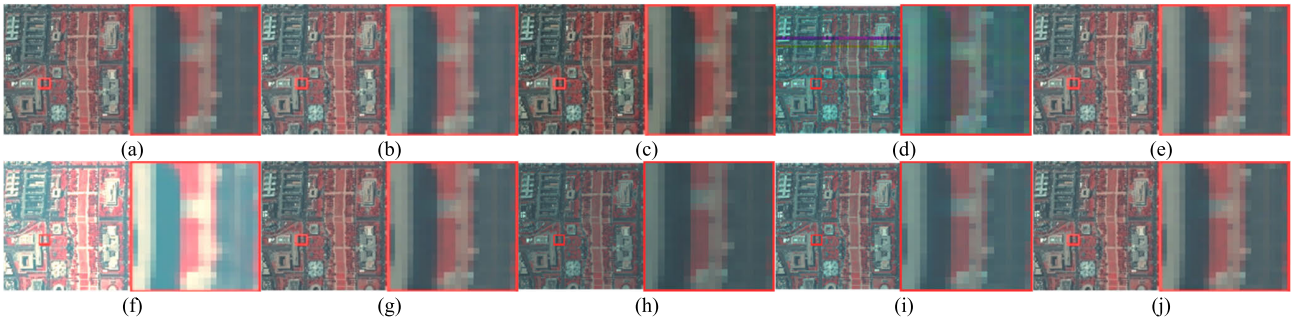


Fig. 14. Results of DC with the Case2 stripes repaired by different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN.

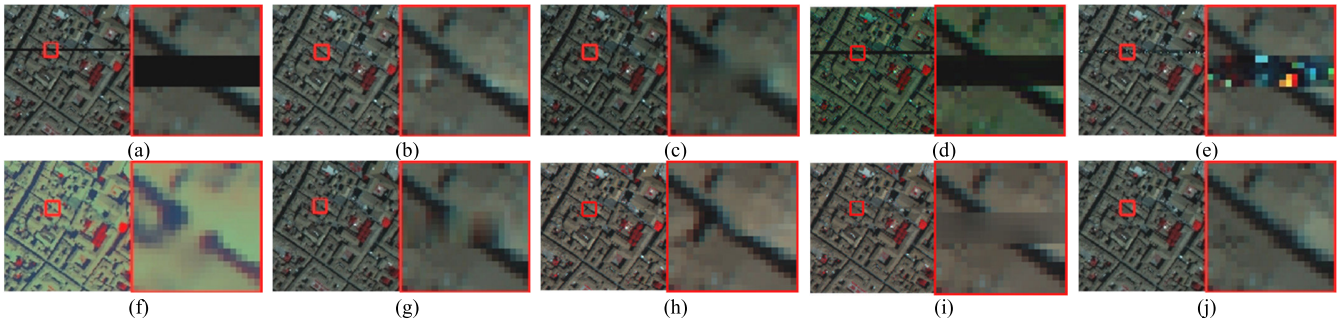


Fig. 15. Results of Pavia with the Case1 stripes repaired by different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN.

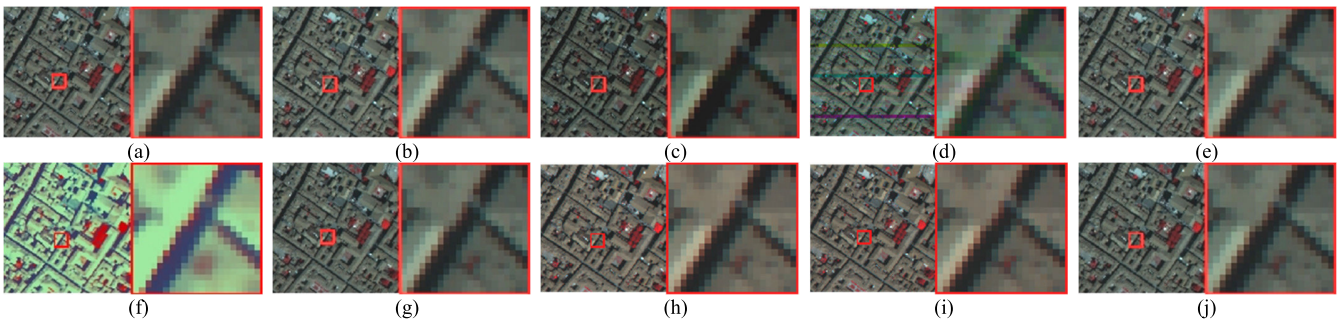


Fig. 16. Results of Pavia with the Case2 stripes repaired by different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN.

In addition, SAPN exceeds SQRN and MAN in terms of PSNR, SAM, and ERGAS and lags slightly behind them in terms of SSIM. From Fig. 20(h) and (i), it can be seen that SQRN and MAN recover the spatial information of stripes poorly, and their heat maps of the integral error mean that more pixels suffer from spectral distortion. There are two reasons for their poor results. One aspect is their lack of adaptation to inconsistent absolute radiance intensities between the normalized damaged and ground-truth images, such as the significant errors in the nondamaged regions of Fig. 20(h). Another aspect

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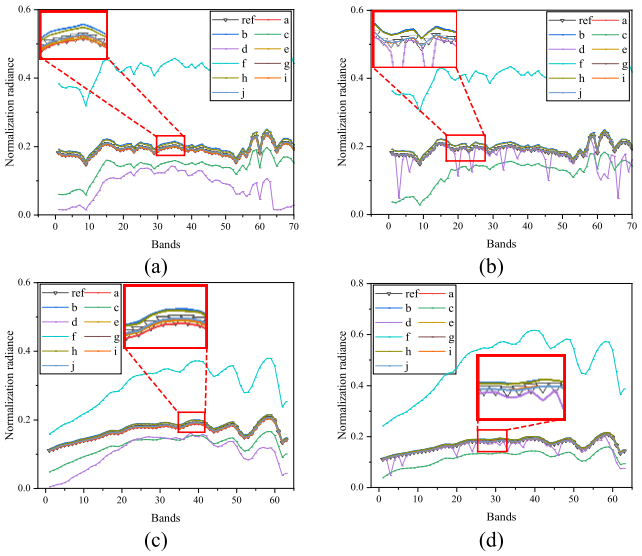


Fig. 17. Spectral curves of DC and Pavia. (a) and (b) Curves of mean normalization radiance of each band in DC with the Case1 and Case2 stripes, respectively, and (c) and (d) those of Pavia with the Case1 and Case2 stripes, respectively. a–j, respectively, represent the methods of FastHyIn, HSDIP, HSIDwRD, GRN, DPHSIR, DeepHyIn, GLIC_NM, RDLRHyIn, GLRTA, and SAPN, and ref represents the ground truth.

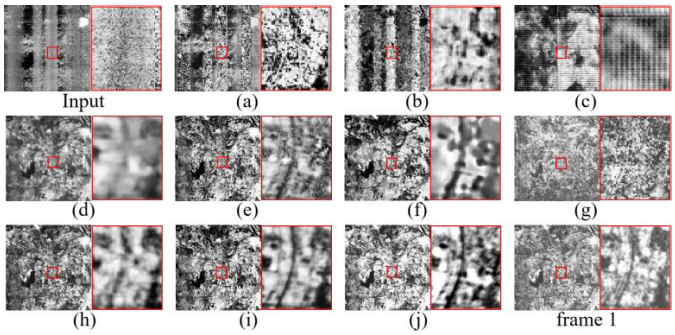


Fig. 18. Results of the 45th band in GF5 repaired by different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN. "frame1" represents the first band of GF5.

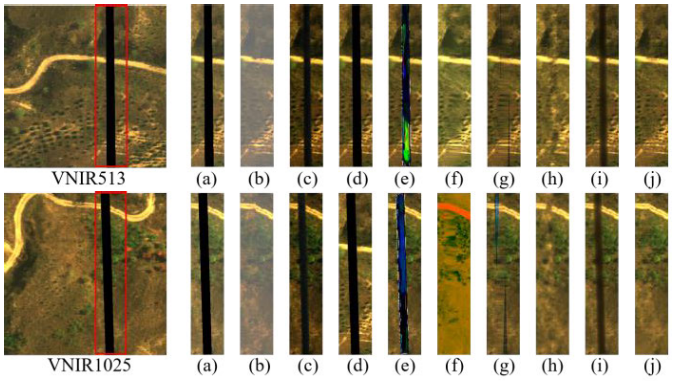


Fig. 19. Results of VNIR513 and VNIR1025 repaired by different methods. (a) FastHyIn. (b) HSDIP. (c) HSIDwRD. (d) GRN. (e) DPHSIR. (f) DeepHyIn. (g) GLIC_NM. (h) RDLRHyIn. (i) GLRTA. (j) SAPN. The red, green, and blue channels of these results correspond to the 29th, 17th, and 8th of the HSI images, respectively.

is the limitations of the algorithm. The attention module used in MAN divides the input features into local blocks and then computes and combines their local attention features, which

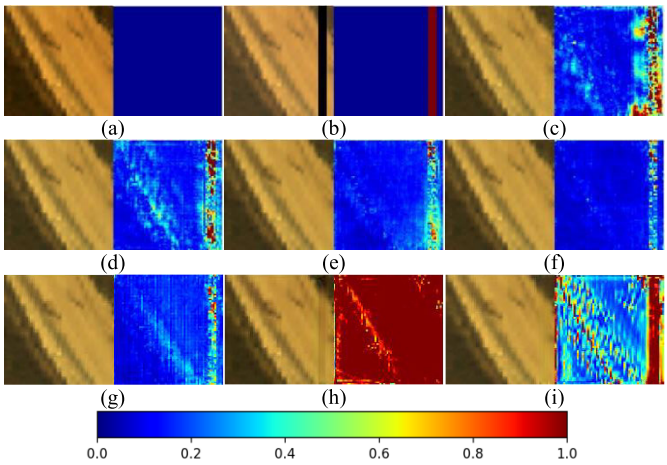


Fig. 20. Visual results of the pruned networks. The left is the RGB images (corresponding to the 29th, 17th, and 8th bands) of 1639.mat, and the right is the heat map of the integral error between the repaired data and the ground truth. (a)–(i) Visual results of ground truth, Case1, Net, Net+P, Net+P+SA (SAPN), Net+P+SMA, Net+P+TSMA, SQRN [43], and MAN [51], respectively. In the heat map, the colors from blue to red represent less to more reconstruction errors. The meanings of Net, P, SA, SMA, and TSMA are introduced in Section IV-F.

weakens the global contextual dependencies of wide-stripe regions. The spectrumwise recursive attention of SQAN fails to handle the Case1 stripe due to the loss of information across all spectral bands. For SAPN, its key module SMA can capture the global dependence of three different planes, and the progressive reconstruction process of SAPN further weakens the negative impact of the lack of information in stripe areas. Therefore, SAPN shows better performance than the attention-based methods of SQRN and MAN.

H. Shareable Model for Unsupervised Restoration

Typically, a deep learning model is trained and then should serve as many images as possible. In order to verify that the proposed unsupervised restoration can obtain an effective shareable model, we train SAPN with different unsupervised learning strategies (including M1–M3) and test them on 50 hyperspectral images from the Case1 dataset. M0 represents the damaged images without restoration operation, M1 means that SAPN adopts the learning strategy of DIP, and its inputs of training and testing are different images, but they are all random noise images that conform to Gaussian distribution and have the same mean and variance. For the M2 strategy, the random noise image for training is saved and then fed to the convergent SAPN for testing. M3 is that SAPN utilizes the proposed unsupervised learning method. M1–M3 all cost 5000 model iteration calculations for each image restoration to ensure a fair comparison. Fig. 21 shows the PSNR results of the restored images. It can be seen that M1 performs the worst, and its results are comparable to those of M0, indicating that the unsupervised learning strategy M1 fails to obtain a shareable model for 50 test images. Compared with M1, M2 improves the PSNR values of the repaired images thanks to the fixed input image. This means that M2 requires additional storage of training inputs and its performance is still unstable and unsatisfactory. M3 obtains the most stable and

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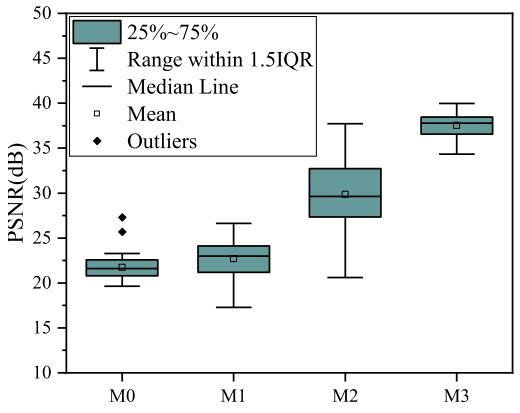


Fig. 21. PSNR results of different unsupervised strategies on 50 images from the Case1 dataset. M0 refers to the damaged images without any restoration, and M1–M3 represent the DIP learning strategy, the DIP learning strategy with the same input, and the proposed unsupervised learning strategy, respectively.

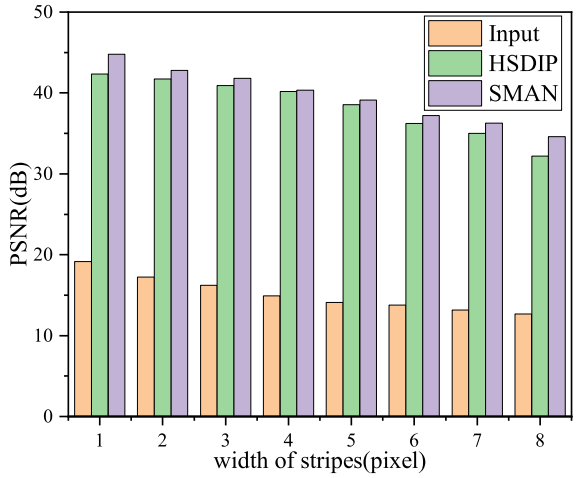


Fig. 22. Inpainting results of images with the Case1 stripes of different widths.

superior performance only with the final converged network, and it proves that the proposed unsupervised restoration can achieve an effective shareable model.

I. Discussion of Stripe Width and Parameters r_c, r_h, and r_w

We perform SAPN on images with stripes of different widths and discuss the impact of r_c, r_h, and r_w in SMA.

For the stripe with different widths, we randomly choose 16 images from the Case1 test dataset and readd the stripes with different widths. The widths of stripes are set as range from 1 to 8 pixels. HSDIP and SAPN all cost 5000 model iteration calculations and use the same training method for fair comparison. Fig. 22 shows the inpainting results. It can be seen that SAPN achieves better PSNR results than HSDIP, and their performance is closest for repairing 4-pixel-wide stripes. Also, SAPN outperforms HSDIP more significantly for narrower or wider stripes. This proves that SAPN has advanced abilities of feature representation and inference for narrower or wider stripes.

Furthermore, we change the setting of $r_c = r_h = r_w = s$ in SMA and retrain and retest SAPN on the Case1 dataset.

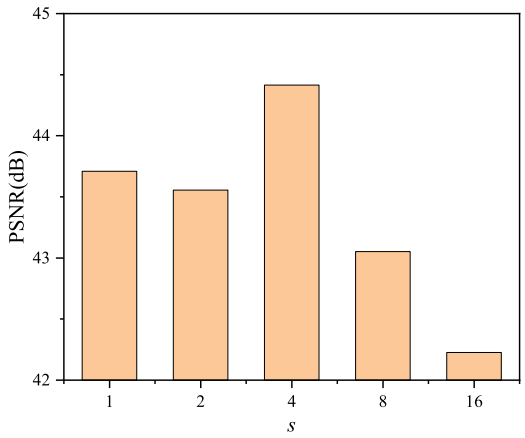


Fig. 23. Results repaired by SAPN with different r_c, r_h, and r_w. "s" represents the setting of $r_c = r_h = r_w = s$.

The constant s is set as 1, 2, 4, 8, and 16. Fig. 23 presents the performance of SAPN with different s's. It can be seen that the setting of $s = 4$ enables SAPN to achieve the best performance. The performance of SAPN decreases as the value of s increases except for $s = 4$. A larger value of s means greater computational complexity, and thus, $s = 4$ is both the optimal setting and a compromise in computational overhead.

V. Conclusion

This article proposes SAPN to restore HSIs with wide stripes. Inspired by the experience that narrow stripes are easy to repair, this article decomposes wide-stripe restoration into narrow-stripe restoration and upsampling and constructs EPB to realize progressive image restoration. In EPB, the coarse-to-fine image restoration of upsampling alleviates the pixel neighborhood loss caused by too wide stripes. Moreover, SMA, as an advanced spatial–spectral attention module, mines multidirectional context priors to weaken the negative impact of missing neighbors, and it enhances the feature inference ability of EPB. Furthermore, this article develops a DIP-based unsupervised restoration strategy, and it introduces the simulated ground truth to supervise SAPN to learn multisample priors, thereby overcoming the problems of DIP in terms of recurrence and overhead. Experiments are performed on private and public datasets where wide stripes exist at the same and different spatial locations. The comparison of different methods reveals the significant advantages of SAPN in terms of objective metrics, spatial texture, and spectral fidelity. In addition, inpainting-based methods are better at wide-stripe restoration compared to denoising-based methods, providing some insight for developing wide-stripe restoration methods.

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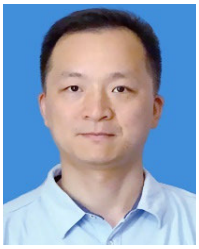
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